

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 4: Solutions

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1. Refer to the electric circuit shown in Figure 1. Evaluate V_{OC} , I_{SC} and R_{Th} .

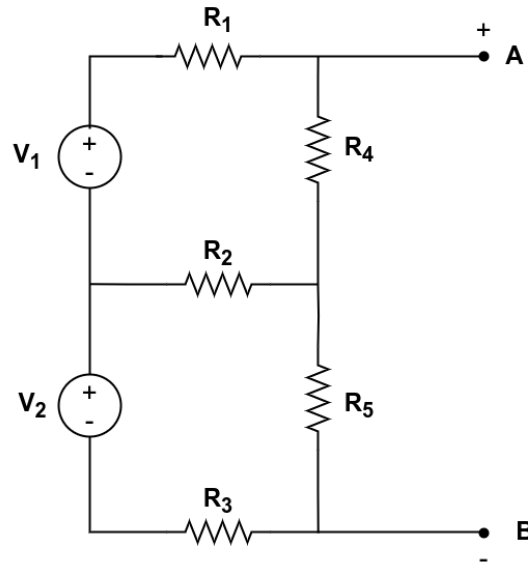


Figure 1 The electric circuit has two independent sources $V_1 = V_2 = 120V$ and the resistances $R_1 = R_2 = R_3 = 0.1\Omega$, $R_4 = 9.6\Omega$ and $R_5 = 5.76\Omega$

Solution:

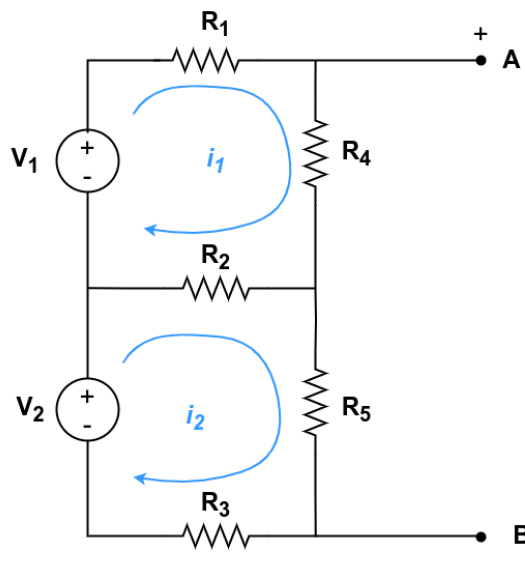


Figure 2 Solution of the mesh currents i_1 and i_2

KVL in mesh-1 gives

$$-V_1 + i_1 R_1 + i_1 R_4 + (i_1 - i_2) R_2 = 0$$

Rearranging, we get

$$(R_1 + R_2 + R_4)i_1 - R_2i_2 = V_1$$

KVL in mesh-2 gives

$$-V_2 + R_5i_2 - (i_1 - i_2)R_2 + R_3i_2 = 0$$

Rearranging, we get

$$-R_2i_1 + (R_2 + R_3 + R_5)i_2 = V_2$$

Thus, in $ZI = DU$ form we get

$$\begin{bmatrix} R_1 + R_2 + R_4 & -R_2 \\ -R_2 & R_2 + R_3 + R_5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Using the numerical values of the circuit elements, we obtain the following solution of the above equations

$$i_1 = 12.452 \text{ A} , i_2 = 20.343 \text{ A}$$

$$V_{OC} = i_1R_4 + i_2R_5 = 236.715 \text{ V}$$

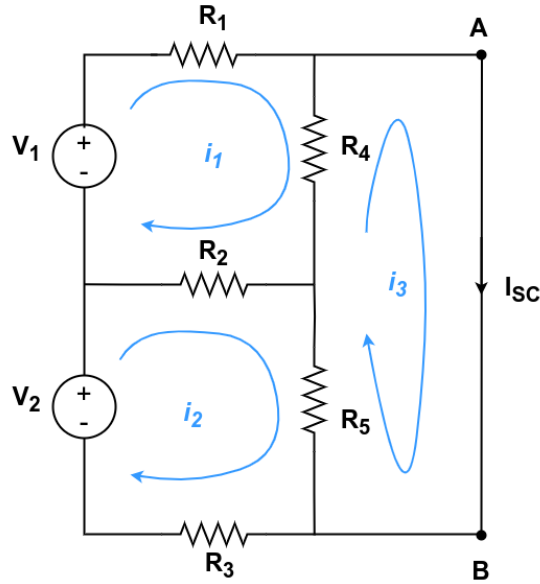


Figure 3 Terminals A and B are short-circuited. The short-circuit current is I_{SC} .

Refer to Figure 3 for the solution of the short-circuit current I_{SC} . Applying KVL in mesh-1, 2 and 3, we have the following equations.

$$(R_1 + R_2 + R_4)i_1 - R_2i_2 - R_4i_3 = V_1 \quad (1)$$

$$-R_2i_1 + (R_2 + R_3 + R_5)i_2 - R_5i_3 = V_2 \quad (2)$$

$$-R_4i_1 - R_5i_2 + (R_4 + R_5)i_3 = 0 \quad (3)$$

Solving the above equations with the numerical values, we get

$$i_1 = i_2 = i_3 = I_{SC} = 1200 \text{ A}$$

Thus,

$$R_{Th} = \frac{V_{OC}}{I_{SC}} = \frac{236.715}{1200} = 0.197\Omega$$

2. Refer to the electric circuit shown in Figure 4. Evaluate V_{OC} , I_{SC} and R_{Th}

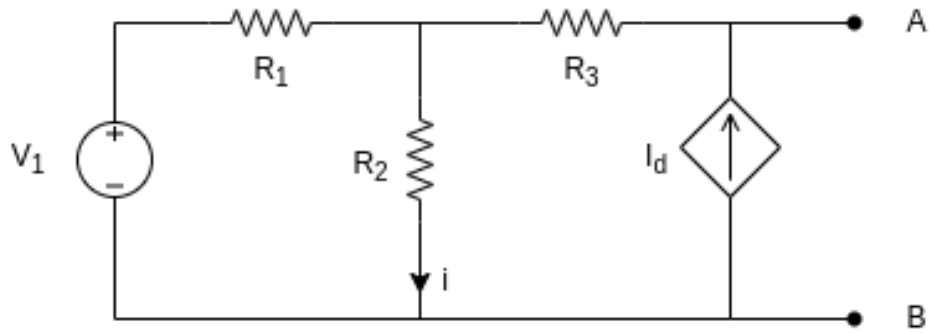


Figure 4 Electric Circuit with independent voltage source $V_1 = 20V$, dependent current source $I_d = ki = 1.5i$ and resistances $R_1 = 40\Omega$, $R_2 = 200\Omega$ and $R_3 = 100\Omega$.

Solution:

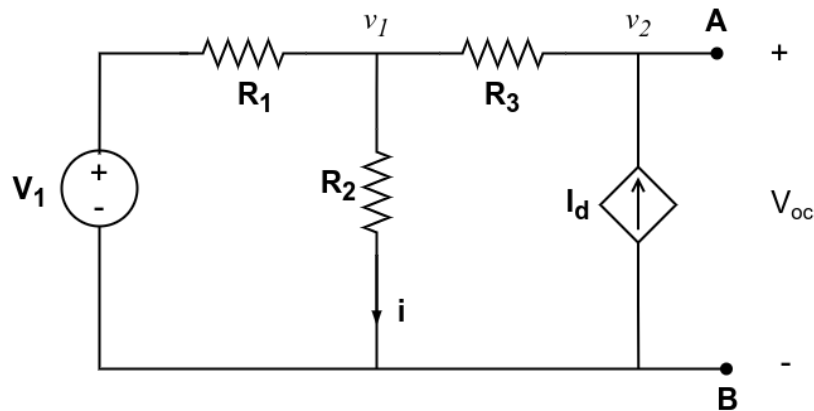


Figure 5 Electric Circuit with independent voltage source $V_1 = 20V$, dependent current source $I_d = ki = 1.5i$ and resistances $R_1 = 40\Omega$, $R_2 = 200\Omega$ and $R_3 = 100\Omega$.

Source transformation of V_1 in series with R_1 to obtain $I_1 = G_1V_1$ in parallel with R_1 .

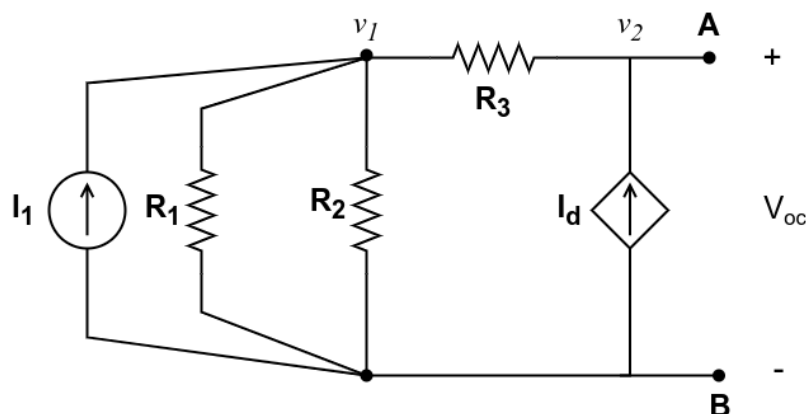


Figure 6 Source transformed version of the circuit shown in Figure: 5.

KCL at Node-1 gives

$$I_1 - G_1v_1 - G_2v_1 - G_3(v_1 - v_2) = 0$$

Rearranging, we get

$$(G_1 + G_2 + G_3)v_1 - G_3v_2 = I_1$$

KCL at Node-2 gives

$$\begin{aligned} G_3(v_1 - v_2) + I_d &= 0 \\ G_3(v_1 - v_2) + kG_2v_1 &= 0 \quad \text{since} \quad [I_d = ki = kG_2v_1] \end{aligned}$$

Rearranging we get

$$-(kG_2 + G_3)v_1 + G_3v_2 = 0$$

Thus in $YV = CU$ form, we have the following

$$\begin{bmatrix} G_1 + G_2 + G_3 & -G_3 \\ -(kG_2 + G_3) & G_3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} [I_1]$$

Solving the above equations using the numerical values provides us with $V_{OC} = v_2 = 38.89V$

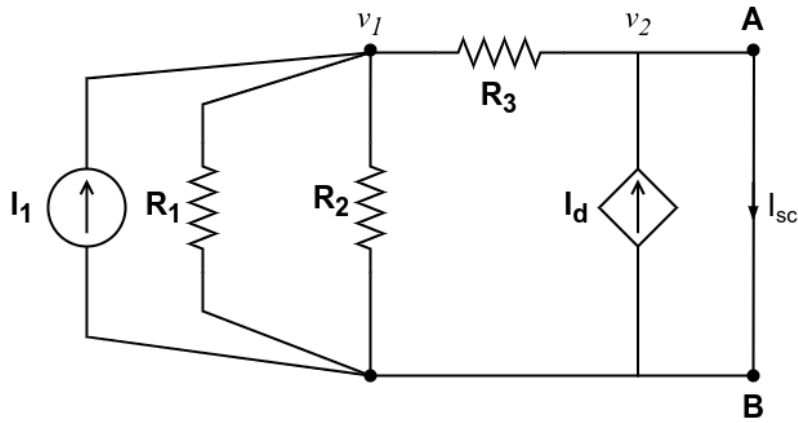


Figure 7 The terminals A and B are short circuited. I_{SC} is the short circuit current from A to B

Refer to Fig.7 where the terminals A and B are shorted. Here Node-2 is connected to the reference Node.

KCL at Node-1 gives

$$\begin{aligned} I_1 - G_1v_1 - G_2v_1 - G_3v_1 &= 0 \\ v_1 &= \frac{I_1}{G_1 + G_2 + G_3} \end{aligned}$$

KCL at Node-2 gives

$$\begin{aligned} G_3v_1 + I_d - I_{sc} &= 0 \\ I_{sc} &= G_3v_1 + kG_2v_1 \end{aligned}$$

Using the numerical values of the circuit elements, we get $I_{SC} = 218.75 \text{ mA}$.

Thus,

$$\begin{aligned} R_{Th} &= \frac{V_{OC}}{I_{SC}} \\ &= 177.783 \Omega \end{aligned}$$

3. Consider the Boolean function

$$F(A, B, C, D) = \sum(1, 3, 4, 5, 6, 9, 11, 12, 14, 15).$$

- Draw the Karnaugh map for the function F .
- Find all Boolean expressions for F in the minimal SOP form.
- Find all Boolean expressions for F in the minimal POS form.

Solution

- The K-map for the function F is

CD \ AB	00	01	11	10
00	0	1	1	0
01	1	1	0	1
11	1	0	1	1
10	0	1	1	0

- In the K-map above, we can consider the coverings of 1's as shown below. It is clear that the

CD \ AB	00	01	11	10
00	0	1	1	0
01	1	1	0	1
11	1	0	1	1
10	0	1	1	0

1 at location 5 and the 1 at location 15 each have two different ways of covering them. Thus the minimal SOP expressions are not unique. They can be written as

$$\overline{B}.D + B.\overline{D} + \overline{A}.B.\overline{C} + A.B.C$$

$$\overline{B}.D + B.\overline{D} + \overline{A}.\overline{C}.D + A.C.D$$

$$\overline{B}.D + B.\overline{D} + \overline{A}.B.\overline{C} + A.C.D$$

$$\overline{B}.D + B.\overline{D} + \overline{A}.\overline{C}.D + A.B.C$$

(c) In the K-map above, we can consider the coverings of 0's as shown below. The unique minimal

CD \ AB	00	01	11	10
00	0	1	1	0
01	1	1	0	1
11	1	0	1	1
10	0	1	1	0

POS expression is $(B + D) \cdot (\bar{A} + \bar{B} + C + \bar{D}) \cdot (A + \bar{B} + \bar{C} + \bar{D})$.

4. Consider the K-map of the function $G(A, B, C, D)$ shown below in the figure below, with don't cares marked with the symbol ϕ . The designer can arbitrarily assign 0 or 1 to each location marked by ϕ (independently of each other). For example, the designer could for convenience choose the locations 2 and 7 to be 1's and choose locations 8 and 13 to be 0's resulting in a particular choice for the function $G(A, B, C, D) = \sum(0, 1, 2, 3, 4, 5, 7, 12)$.

- (a) Find all Boolean expressions for G in the minimal SOP form.
 (b) Find all Boolean expressions for G in the minimal POS form.

CD \ AB	00	01	11	10
00	1	1	1	ϕ
01	1	1	ϕ	0
11	1	ϕ	0	0
10	ϕ	0	0	0

Solution

The following line of reasoning leads to the solution.

- (a) For the SOP, we need to cover all the 1's in the K-map. Four of them can be covered by one product term as shown below.

CD \ AB	00	01	11	10
00	1	1	1	ϕ
01	1	1	ϕ	0
11	1	ϕ	0	0
10	ϕ	0	0	0

This leaves the 1's at locations 3 and 12 that needs to be covered by product terms. The possible choices can be seen to the ones shown below.

CD \ AB	00	01	11	10
00	1	1	1	1
01	1	1	0	0
11	1	0	0	0
10	1	0	0	0

CD \ AB	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	1	1	0	0
10	0	0	0	0

The first one corresponds to the choice of $G(A, B, C, D) = \sum(0, 1, 2, 3, 4, 5, 8, 12)$ with the minimal SOP expression $\overline{A}.\overline{C} + \overline{A}.B + \overline{C}.D$. The second one corresponds to $G(A, B, C, D) = \sum(0, 1, 3, 4, 5, 7, 12, 13)$ with the minimal SOP expression $\overline{A}.\overline{C} + \overline{A}.D + \overline{C}.B$.

- (b) Similarly, for the POS, we need to cover all the 0's in the K-map. Four of them can be covered by one product term as shown below.

CD \ AB	00	01	11	10
00	1	1	1	ϕ
01	1	1	ϕ	0
11	1	ϕ	0	0
10	ϕ	0	0	0

This leaves the 0's at locations 6 and 9 that needs to be covered by sum terms. The possible choices can be seen to the ones shown below.

CD \ AB	00	01	11	10
00	1	1	1	0
01	1	1	1	0
11	1	1	0	0
10	0	0	0	0

CD \ AB	00	01	11	10
00	1	1	1	1
01	1	1	0	0
11	1	0	0	0
10	1	0	0	0

The first one corresponds to the choice of $G(A, B, C, D) = \prod(2, 6, 8, 9, 10, 11, 14, 15)$ with the minimal POS expression $(\overline{A} + \overline{C}).(\overline{A} + B).(\overline{C} + D)$. The second one corresponds to $G(A, B, C, D) = \prod(6, 7, 9, 10, 11, 13, 14, 15)$ with the minimal POS expression $(\overline{A} + \overline{C}).(\overline{A} + \overline{D}).(\overline{C} + \overline{B})$.